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CUET UG

Chemistry

Aldehydes, Ketones and Carboxylic Acids

Official Syllabus Unit VIII · Class 12 · CUET UG 2027 Cycle Notes

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Aligned to the latest official syllabus · Concept-first explanations

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Syllabus Alignment

Sub-topic	Official syllabus pointer (NTA CUET(UG) Chemistry syllabus 2026, the current 2027 baseline, Unit VIII)	Skills the questions test
Carbonyl family, names and structure	General introduction to carbonyl compounds, Nomenclature of Aldehydes, Ketones and carboxylic acids, structure of Carbonyl group	Writing -al, -one and -oic acid names, explaining the sp^2 polar C=O, and ranking aldehyde vs ketone reactivity from steric and electronic grounds.
Preparing aldehydes and ketones	Methods of preparation of Aldehydes & Ketones	Selecting among alcohol oxidation, Rosenmund, Etard, Gattermann-Koch, acyl-chloride + cadmium reagent, nitrile routes, ozonolysis and alkyne hydration.
Carbonyl properties and reactions	Physical properties of aldehydes & ketones (boiling point & solubility), Chemical properties of aldehydes & ketones, Uses of Aldehydes & Ketones	Predicting nucleophilic additions, reductions (Clemmensen, Wolff-Kishner), the Tollen's/ Fehling's/iodoform tests, aldol vs Cannizzaro, and everyday uses like formalin and acetone.
The carboxyl group	Structure of Carboxyl group, Methods of preparation of Carboxylic Acids, Physical properties (Boiling point & Solubility) and Chemical properties of carboxylic acids, Uses of Carboxylic Acids	Explaining acidity and its substituent trends, choosing preparations (oxidation, nitrile hydrolysis, Grignard + CO_2, HVZ), and the derivative conversions — ester, chloride, anhydride, amide — plus decarboxylation.

EXAM PATTERN

CUET UG pattern (NTA official information bulletin and syllabus, 2026 cycle — the current 2027 baseline): each subject paper carries **50 compulsory questions** in **60 minutes**, with +5 for every correct answer and −1 for every incorrect answer. The Chemistry paper maps strictly to NCERT Class 12, and Unit VIII carries the densest named-reaction list in organic chemistry — Rosenmund, Clemmensen, HVZ, Kolbe — where recognising the reagent usually IS the answer.

The Carbonyl Family: One Bond, Three Titles

The C=O group is organic chemistry's busiest junction — it sits in sugars, formalin, nail-polish remover and vinegar. When the carbonyl carbon carries **at least one hydrogen**, the compound is an **aldehyde**; when it carries **two carbons**, a **ketone**; and when an OH joins the carbonyl carbon, you have the **carboxylic acid** — a hybrid whose OH changes the rules entirely.

- **Structure:** the carbonyl carbon is sp^2-hybridised and planar; the C=O stretches about 122 pm — shorter and stronger than C–O, and **strongly polar** (O pulls the π cloud). The carbon becomes the electrophilic target; the oxygen the nucleophilic end.
- **Aldehyde vs ketone reactivity:** aldehydes attack **faster** — one hydrogen exposes the carbon, while ketones carry two alkyl groups that both crowd the attacker (+I push) and feed electron density back. Order: HCHO > aldehydes > ketones.
- **Names:** -al (ethanal, benzaldehyde — always C-1, no locant needed), -one (propanone, pentan-3-one — locant required), -oic acid (ethanoic acid). Aromatic members: benzaldehyde, acetophenone, benzoic acid.
- **The acid's shape:** two resonance forms blend the two C–O bonds to identical length (125 pm each, halfway between single and double) — the carboxylate's symmetry is the physical footprint of the acidity you will meet below.

Preparing Aldehydes and Ketones

Aldehydes — the art of stopping halfway

- **From 1° alcohols:** controlled oxidation — PCC (pyridinium chlorochromate) stops exactly at the aldehyde; $K_2Cr_2O_7$ would overshoot to the acid.
- **Rosenmund reduction:** acid chloride + H_2 over Pd supported on $BaSO_4$ (the $BaSO_4$ **poisons** the catalyst, preventing over-reduction to alcohol) → RCHO. The formaldehyde exception: HCHO cannot be made this way (formyl chloride does not exist) — it comes from methanol oxidation.
- **Etard reaction:** toluene + chromyl chloride (CrO_2Cl_2) → benzaldehyde (the hydrolysis releases the aldehyde).
- **Gattermann-Koch:** benzene + CO + HCl over anhydrous $AlCl_3$/CuCl → benzaldehyde — the formyl group installed straight onto the ring.
- **Side-chain oxidation:** toluene + CrO_2Cl_2; careful — strong oxidants ($KMnO_4$) jump straight to benzoic acid.

Ketones — fair game for strong oxidants

- **From 2° alcohols:** oxidation with $K_2Cr_2 \xrightarrow[H^+]{O_7}$ — ketones resist further oxidation, so the reaction **stops** cleanly (unlike aldehydes).
- **From acid chlorides:** $R\text{–}COCl + R'_2\ Cd \rightarrow R\text{–}CO\text{–}\ R' + R'CdCl$ — dialkylcadmium (from the Grignard + $CdCl_2$) is gentle: it never overshoots to the alcohol.
- **From nitriles:** R–CN + Grignard, then hydrolysis → ketone (Stevens? — simply nitrile + RMgX).
- **Ozonolysis of alkenes:** cleaves C=C and carbonyls pop out where the double bond sat.
- **Alkynes + water:** $CH \equiv CH + H_2O$ over $\xrightarrow[H_2]{HgSO_4} SO_4$ → ethanal (Markovnikov addition, tautomerisation); terminal alkynes give methyl ketones.
- **From benzenoid rings:** Friedel-Crafts acylation — benzene + $RCOCl/AlCl_3$ → aryl ketone (acetophenone).

Physical Properties: Polarity Without H-Bonding

- **Boiling points** slot between alkanes and alcohols: the C=O dipole attracts molecules to each other, but only the alcohol's O–H can hydrogen-bond **to itself**. propan-1-ol (97°C) beats propanal (49°C) beats propane (−42°C).
- **Solubility:** small carbonyls (methanal, ethanal, propanone) mix fully with water — the oxygen accepts H-bonds from water; solubility dives as chains grow.
- **Everyday smells:** vanillin and cinnamaldehyde sweeten; acetone cleans nail polish; formalin (40% formaldehyde) preserves specimens.

Reactions of Aldehydes and Ketones

Nucleophilic addition — the C=O handshake

- **HCN:** adds to give **cyanohydrins** — the carbon chain grows by one; base catalysis (CN^- attack) speeds it.
- **$NaHSO_3$:** forms crystalline **bisulphite adducts** — used to purify and separate aldehydes and methyl ketones (the crystals decompose back on acid/base).
- **Grignard:** R–MgX + HCHO → 1° alcohol; + aldehyde → 2°; + ketone → 3° (Unit VII's chain-builder).
- **Alcohols:** one ROH gives a **hemiacetal**; two give an **acetal** (acid-catalysed, water removed) — sugar chemistry runs on this.
- **Ammonia derivatives:** $NH_2\ OH$ → oxime; NH_2NH_2 → hydrazone; 2,4-DNP → the yellow-orange hydrazone (the test). All proceed via addition then water loss — **addition-elimination**.

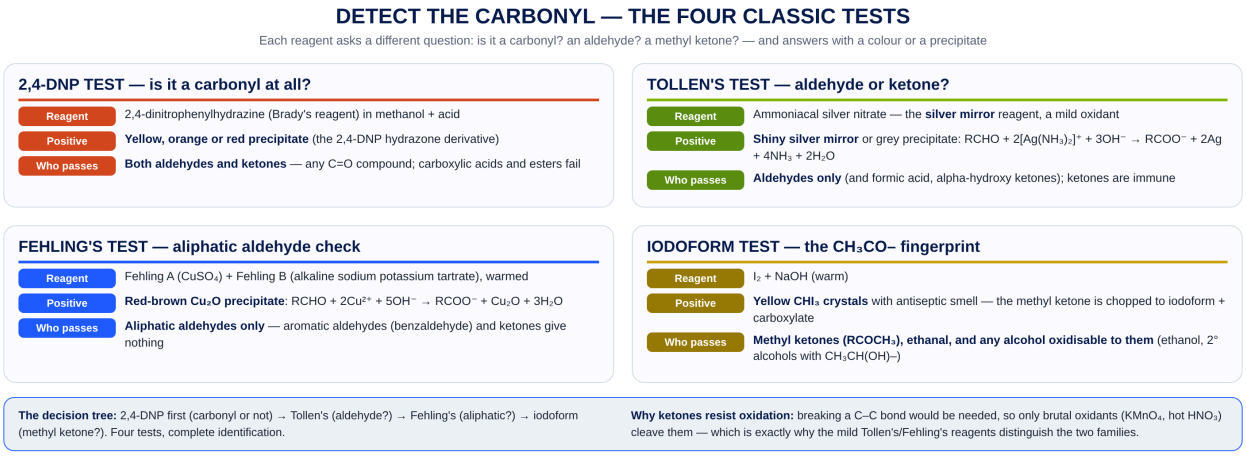


Figure 1: The four classic carbonyl tests — 2,4-DNP, Tollen's, Fehling's and iodoform — with reagents, positive results and the exact sets of compounds that pass each.

Reduction and oxidation

- **Reduction to alcohols:** $NaBH_4$ (gentle) or $LiAlH_4$ (powerful) → aldehydes give 1°, ketones give 2° alcohols.
- **Clemmensen reduction:** C=O → CH_2 using Zn–Hg amalgam + concentrated HCl — for acid-sensitive-free substrates; **Wolff-Kishner:** same goal with N_2H_4 then KOH/ethylene glycol — for base-stable ones. Both strip the oxygen from the carbonyl carbon entirely.
- **Oxidation — the family divider:** aldehydes surrender easily (Tollen's silver mirror, Fehling's red-brown Cu_2O) because the C–H on the carbonyl carbon leaves; ketones **refuse** mild oxidants — only brutal $KMnO_4$ cleaves them, splitting the chain at the C=O.
- **Iodoform reaction:** methyl ketones ($RCOCH_3$) + $\frac{I_2}{NaOH}$ → yellow CHI_3 + carboxylate. Ethanal and ethanol (oxidised in situ to ethanal) also qualify — the $CH_3CO–$ or $CH_3CH(OH)–$ fingerprint.

The α-carbon stories: aldol and Cannizzaro

- **Aldol condensation:** aldehydes/ketones with an α-hydrogen + dilute NaOH → the β-hydroxy "aldol", which heats to the α,β-unsaturated carbonyl. Two acetaldehyde molecules → 3-hydroxybutanal → but-2-enal.
- **Cross-aldol:** two **different** carbonyls, both bearing α-H, give a mixture of four products — predictable but messy.
- **Cannizzaro reaction:** carbonyls with **no α-H** (HCHO, benzaldehyde) instead **disproportionate** with concentrated NaOH: one molecule reduces to the alcohol while the other oxidises to the acid — 2 HCHO + NaOH → CH_3 OH + HCOONa. Aldol needs α-H; Cannizzaro needs the absence — the two never share a substrate.

Carboxylic Acids: The Strongest Organic Acids

Structure and acidity

- **Why acids are acids:** the carboxylate ion left after losing H^+ spreads its charge equally over **two** oxygens — resonance symmetry that neither phenoxide nor alkoxide enjoys. Hence RCOOH > phenol > water > alcohols in acidity.

- **Substituent tuning:** electron-withdrawers raise acidity — $ClCH_2\ COOH < Cl_2\ CHCOOH < Cl_3\ CCOOH$; down the halogens F > Cl > Br > I; closer substituents count more (α > β); electron-donating alkyl groups **lower** it. Benzoic acid edges out acetic (the sp^2 ring pulls).
- **Dimerisation:** two acid molecules hydrogen-bond in pairs — which is why acids boil absurdly high (ethanoic acid 118°C) and benzoic acid sublimes.

Preparations

- **Oxidation:** 1° alcohols and aldehydes → acids ($KMnO_4$, $K_2Cr_2O_7$); ketones only under brutal cleavage.
- **Nitrile hydrolysis:** R–CN + $H_2\frac{O}{H^+}$ (or alkali, then acid) → RCOOH — chain length preserved.
- **Grignard + CO_2:** RMgX poured onto dry ice, then acid → the acid with **one extra carbon**.
- **HVZ reaction:** acids with red P + X_2 halogenate the α-position — $CH_3\ COOH \rightarrow ClCH_2\ COOH$ (needed for the acidity ladder above).
- **Ester/amide hydrolysis:** the reverse of derivative formation.

Reactions

- **Neutralisations:** NaOH → salt; $NaHCO_3$ / Na_2CO_3 → salt + CO_2 fizz — **the** test that separates acids from phenol.
- **Derivatives:** + alcohol/acid → ester (the alcohol's C–O stays whole); + $SOCl_2$ → acid chloride; + P_2O_5 (dehydrate) → anhydride; + NH_3, heat → amide.
- **Decarboxylation:** soda lime (NaOH + CaO, heat) chops the COOH off as CO_2, shortening the chain by one — $CH_3\ COONa + NaOH \rightarrow CH_4 + Na_2CO_3$.
- **Reduction:** $LiAlH_4$ → 1° alcohol (acids are stubborn — borane is the gentle alternative).
- **Halogenation:** only the α-position (HVZ), never the ring — the COOH is meta-directing and deactivating for electrophiles.
- **Uses:** acetic acid in vinegar and aspirin chemistry; methanoic acid in leather; benzoic acid salts as food preservatives; fatty acids as soaps' parents.

Key Reactions & Rules: One-Stop Revision

PREPARATIONS, TESTS, NAMED REACTIONS

Rule / route	Symbols in plain words	When to reach for it
PCC on 1° alcohol	stops at aldehyde	Clean RCHO preparation
Rosenmund: $RCOCl + H_2$, Pd-$BaSO_4$	catalyst poison prevents over-reduction; HCHO impossible	Acid chloride → aldehyde

Rule / route	Symbols in plain words	When to reach for it
Etard: toluene + CrO_2Cl_2	side-chain oxidation to CHO	Benzaldehyde from toluene
Gattermann-Koch: CO + HCl, $AlCl_3$	formyl onto benzene	Ring formylation
R_2 Cd + acid chloride	stops at ketone — gentler than Grignard	Acid chloride → ketone
Alkyne + H_2O, $HgSO_4$	Markovnikov, tautomerise; terminal → methyl ketone	Ketone from alkyne
2,4-DNP	yellow/orange ppt — any carbonyl	First identification test
Tollen's	silver mirror — aldehydes (+HCOOH) only	Aldehyde vs ketone
Fehling's	red Cu_2O — aliphatic aldehydes only	Aromatic aldehyde fails it
Iodoform: $\frac{I_2}{NaOH}$	yellow CHI_3 — methyl ketones, ethanal, ethanol	CH_3CO– fingerprint
Clemmensen: Zn-Hg/HCl	C=O → CH_2 (acidic)	vs Wolff-Kishner (basic, N_2H_4)
Aldol: dilute NaOH	needs α-H; cross-aldol = 4 products	β-hydroxy carbonyl synthesis
Cannizzaro: conc. NaOH	needs NO α-H — disproportionation	HCHO, benzaldehyde only
Acidity ladder	RCOOH > phenol > water > alcohol; EWG up, alkyl down; F > Cl > Br > I	pKa ordering questions
$NaHCO_3$ fizz	acid + bicarbonate → CO_2; phenol silent	Acid vs phenol test
HVZ: X_2/red P	α-halogenation of acids	Making $ClCH_2COOH$
Soda lime	decarboxylation — chain shrinks by 1	CH_3COONa → CH_4

Exceptions & Traps

THE TRAPS THIS CHAPTER SETS — READ BEFORE THE MOCK

The rule you assume	The exception / twist	The trap in the question
Every aldehyde passes every aldehyde test	Fehling's rejects aromatic aldehydes — benzaldehyde stays blue	Tollen's vs Fehling's discrimination questions
Grignard + acid chloride gives the alcohol	Dialkylcadmium stops at the ketone; plain RMgX overshoots	"Why cadmium reagent" — its gentleness is the whole point
All carbonyls give aldol	No α-H means NO aldol — Cannizzaro instead	HCHO/benzaldehyde options labelled "aldol product" are bait
Rosenmund works for every aldehyde	HCHO cannot form (no formyl chloride)	The "Rosenmund for formaldehyde" trap
Carboxylic acid + Grignard = alcohol	Acids DESTROY Grignards (proton transfer) — use CO_2 with the reagent first	The dry-ice synthesis order question
Halogenation puts X anywhere	HVZ allows only the α-position, and needs red P	Ring vs α-position product options
Ketones never oxidise	Mild reagents no; $KMnO_4$/ hot HNO_3 cleave C–C	"Oxidation product of butanone" chain-cleavage questions
HCOOH is just an acid	It carries an aldehyde-like H — passes Tollen's AND Fehling's	The formic-acid double-identity MCQ

The Whole Chapter in One Picture

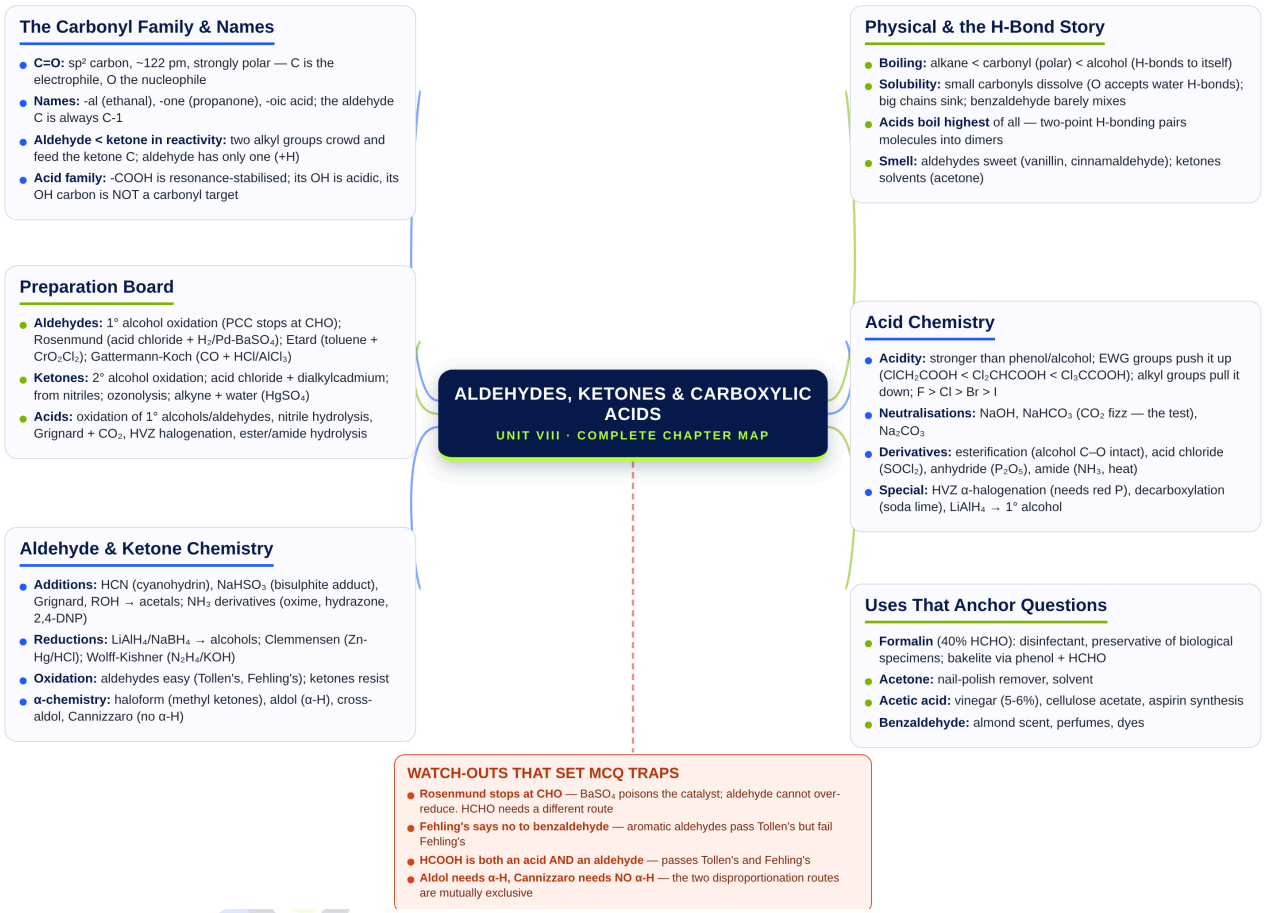


Figure 2: Chapter mindmap: the carbonyl family and its names, the preparation board, physical trends, aldehyde/ketone reactions with the four tests, acid chemistry and derivative conversions, and signature uses — with the traps that catch most students.

Practice Questions (Exam Pattern)

Ten questions in the CUET UG style. Attempt all of them — the paper gives +5 for each correct answer and −1 for every incorrect answer.

Q1. Which compound gives a positive Tollen's test but fails Fehling's test?

(A) ethanal
(B) propanal
(C) benzaldehyde
(D) propanone

Q2. Rosenmund reduction stops at the aldehyde because:

(A) aldehydes resist reduction
(B) $BaSO_4$ poisons the Pd catalyst, preventing over-reduction
(C) H_2 is weak
(D) the acid chloride is unreactive

Q3. Which pair undergoes the Cannizzaro reaction?

(A) ethanal and propanal
(B) formaldehyde and benzaldehyde
(C) acetone and butanone
(D) acetaldehyde and benzaldehyde

Q4. The iodoform test is positive for:

(A) benzophenone
(B) acetone
(C) methanol
(D) propanal

Q5. Dialkylcadmium reacts with an acid chloride to give mainly:

(A) a tertiary alcohol
(B) a ketone
(C) an aldehyde
(D) an ester

Q6. The correct acidity order is:

(A) $HCOOH > CH_3\ COOH$
(B) $CH_3\ COOH > HCOOH$
(C) $ClCH_2\ COOH < CH_3\ COOH$
(D) phenol > acetic acid

Q7. Carboxylic acids boil far higher than alcohols of similar mass because:

(A) they are heavier
(B) two acid molecules H-bond into dimers
(C) the COOH is polar only
(D) they ionise completely

Q8. HVZ halogenation of acetic acid (X_2, red P) gives:

(A) ring-brominated acid
(B) $ClCH_2\ COOH$ (α-substitution)
(C) decarboxylated product
(D) an ester

Q9. Sodium salt of an acid with soda lime gives:

(A) an alcohol
(B) an alkane (chain shortened by one)
(C) an alkene
(D) an amide

Q10. Benzene + CO + HCl over $AlCl_3$ (Gattermann-Koch) gives:

(A) benzyl alcohol
(B) benzaldehyde
(C) benzoic acid
(D) acetophenone

Detailed Solutions

Sol. Q1. Tollen's oxidises ALL aldehydes; Fehling's only the **aliphatic** ones — benzaldehyde's ring keeps the carbonyl carbon too stable. Benzaldehyde: silver yes, red Cu_2O no.

Answer: (C)

Sol. Q2. The $BaSO_4$ support deliberately poisons Pd — mild enough to stop at RCHO. Unpoisoned catalysts would run to the alcohol. And HCHO is impossible here: formyl chloride does not exist.

Answer: (B)

Sol. Q3. Cannizzaro demands NO α-hydrogen. Formaldehyde and benzaldehyde both lack one, so concentrated NaOH disproportionates them — one molecule to alcohol, one to acid. Ethanal (α-H) would aldol instead.

Answer: (B)

Sol. Q4. Iodoform needs $CH_3CO–$ (or $CH_3CH(OH)–$). Acetone is the classic methyl ketone → yellow CHI_3. Benzophenone lacks the methyl; methanol fails; propanal is an aldehyde without the methyl-ketone motif.

Answer: (B)

Sol. Q5. The cadmium reagent is gentler than a Grignard — it adds once to the acid chloride and stops at the ketone, never reaching the alcohol. That selectivity is its entire job.

Answer: (B)

Sol. Q6. HCOOH beats CH_3COOH — no electron-pushing alkyl; and $ClCH_2$ COOH beats acetic (the Cl pulls). Phenol sits far below acetic. Only A orders two true members correctly.

Answer: (A)

Sol. Q7. Two COOH molecules pair through two hydrogen bonds — a cyclic dimer — effectively doubling the molecular mass in the liquid. Ethanoic acid boils at 118 °C while its dimer partner logic beats propanol's 97 °C.

Answer: (B)

Sol. Q8. HVZ (Hell-Volhard-Zelinsky) halogenates ONLY the α-carbon, and red P is the required catalyst converting the acid to its acyl halide first. Ring halogenation never happens — COOH is too deactivating.

Answer: (B)

Sol. Q9. Soda lime (NaOH/CaO, heat) decarboxylates: CH_3 $COONa + NaOH \rightarrow CH_4 + Na_2CO_3$ — the classic chain-shortening step in conversion sequences.

Answer: (B)

Sol. Q10. Gattermann-Koch installs a formyl group (–CHO) straight onto the ring: benzaldehyde. Acetophenone needs Friedel-Crafts **acylation** with CH_3 COCl instead.

Answer: (B)

Wrong-Answer Clinic — the one-line corrections

Dropped marks above? Find your slip here and rewrite the corrected rule in your own words.

If you answered...	Your slip was...	The corrected rule, in one line
Q1 option A or B	Treating Fehling's as universal	Fehling's is aliphatic-only; aromatic aldehydes pass Tollen's but fail Fehling's
Q3 option A	Ignoring the α-H requirement	Aldol needs α-H; Cannizzaro needs its absence — check the carbon next to C=O first
Q4 option A	Reading "ketone" as automatic	The test smells the $CH_3CO–$ group — benzophenone carries no methyl
Q6 option C	Trusting "alkyl strengthens acid"	Alkyl groups push electrons and WEAKEN acids; halogens strengthen them
Q10 option C	Confusing formyl with carboxyl	Gattermann-Koch delivers –CHO; benzoic acid needs actual oxidation of the side chain

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